Modernising a Proof Assistant

Creating a competitor to Lean!?

`isabelle-revamp`
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Alex Hyman

Proof assistants are *so in right now*

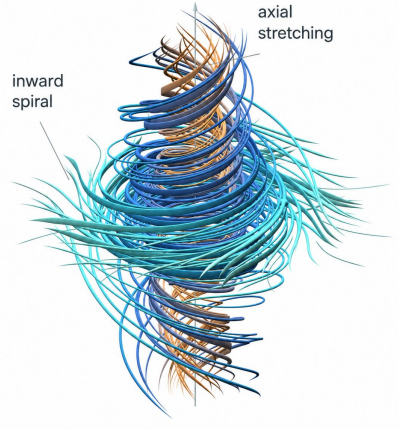
- Proof Assistants + LLMs $\longrightarrow$ Navier Stokes Problem Blown Up!



- Lean is the *de facto* proof assistant for AI labs.

- We ought to have more than one — competition is healthy!

LEAN
THEOREM PROVER

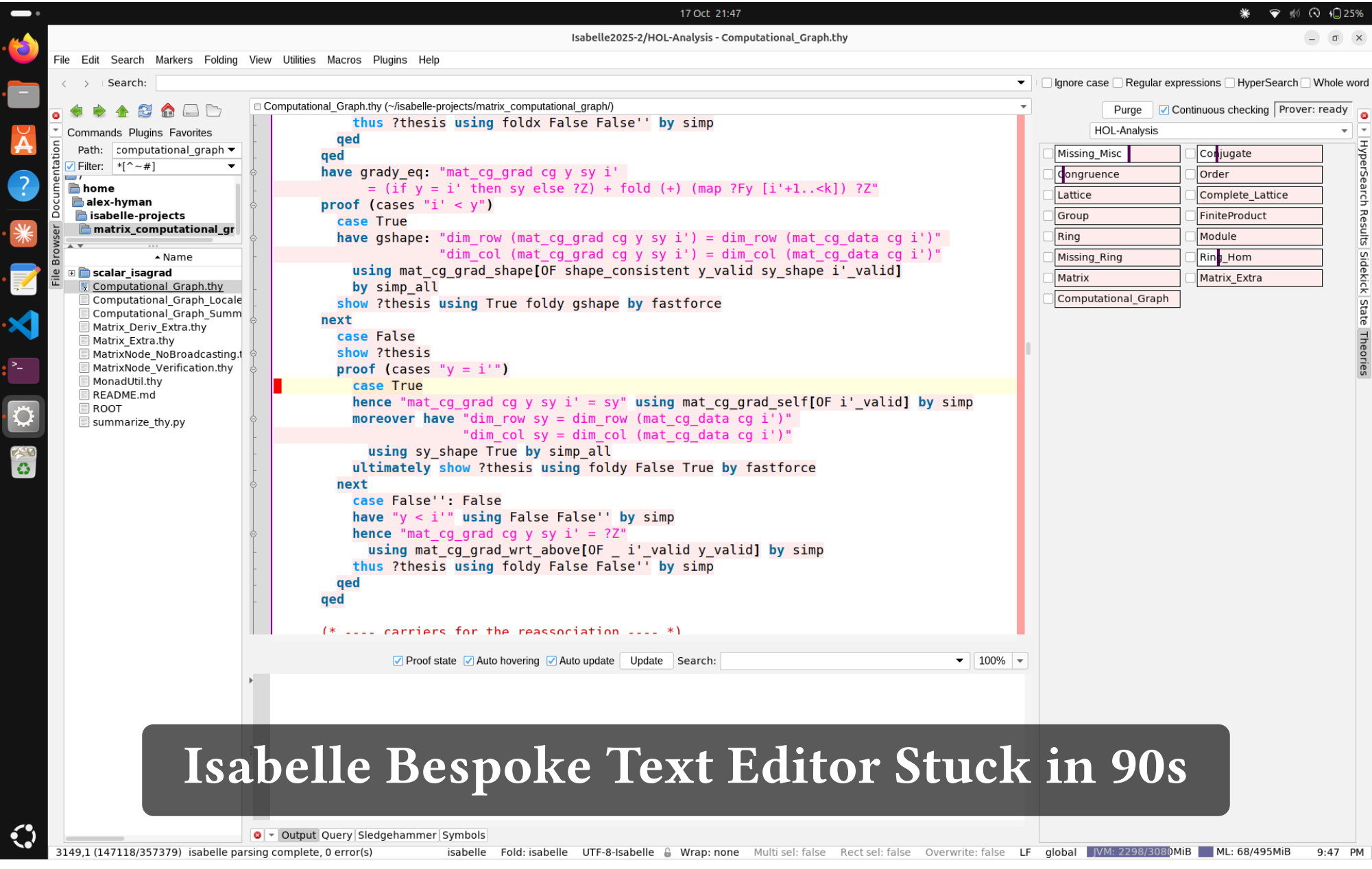


Making Isabelle Fit for the Era of AI



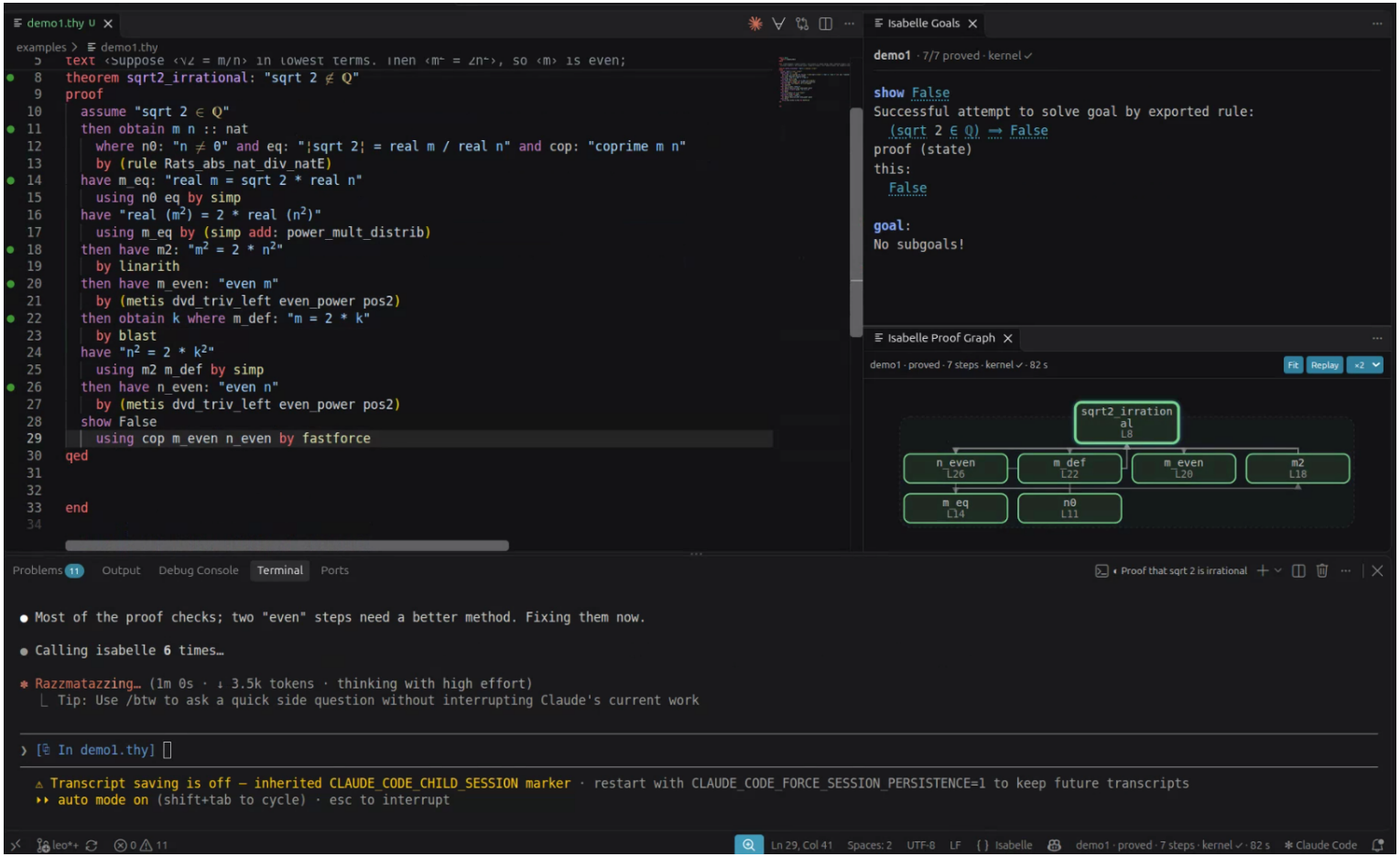
Isabelle *was* the GOAT, but is now **unc**:

- No native VS Code Integration!? ❌
- Little-to-no agentic support ❌
- Intimidating and difficult software installation + setup ❌
- Mature, extensive maths and computer science library ✅
- Strongest (traditional) automation of all proof assistants ✅
- Trustworthy, battle-tested (3 decades) kernel for sound reasoning ✅
- Legible, intuitive proof syntax > Lean's ✅



Isabelle Bespoke Text Editor Stuck in 90s

Demo video



youtu.be/W3U83yHAe08 Video at 2x Speed

I Can't Believe This Didn't Exist Before!

As a PhD proof assistant researcher I think this tool is awesome. Big thanks to Leo and Ark for making it happen!

And if you don't trust me, take it from the man whose PhD supervisor created Isabelle:

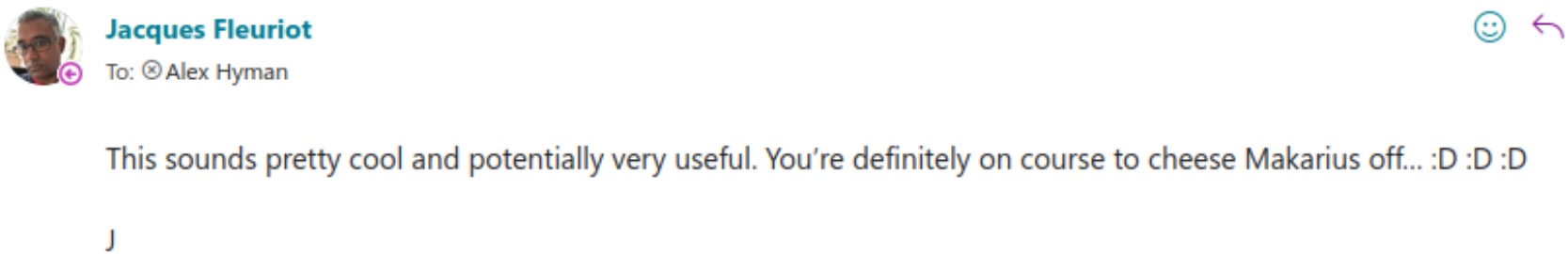
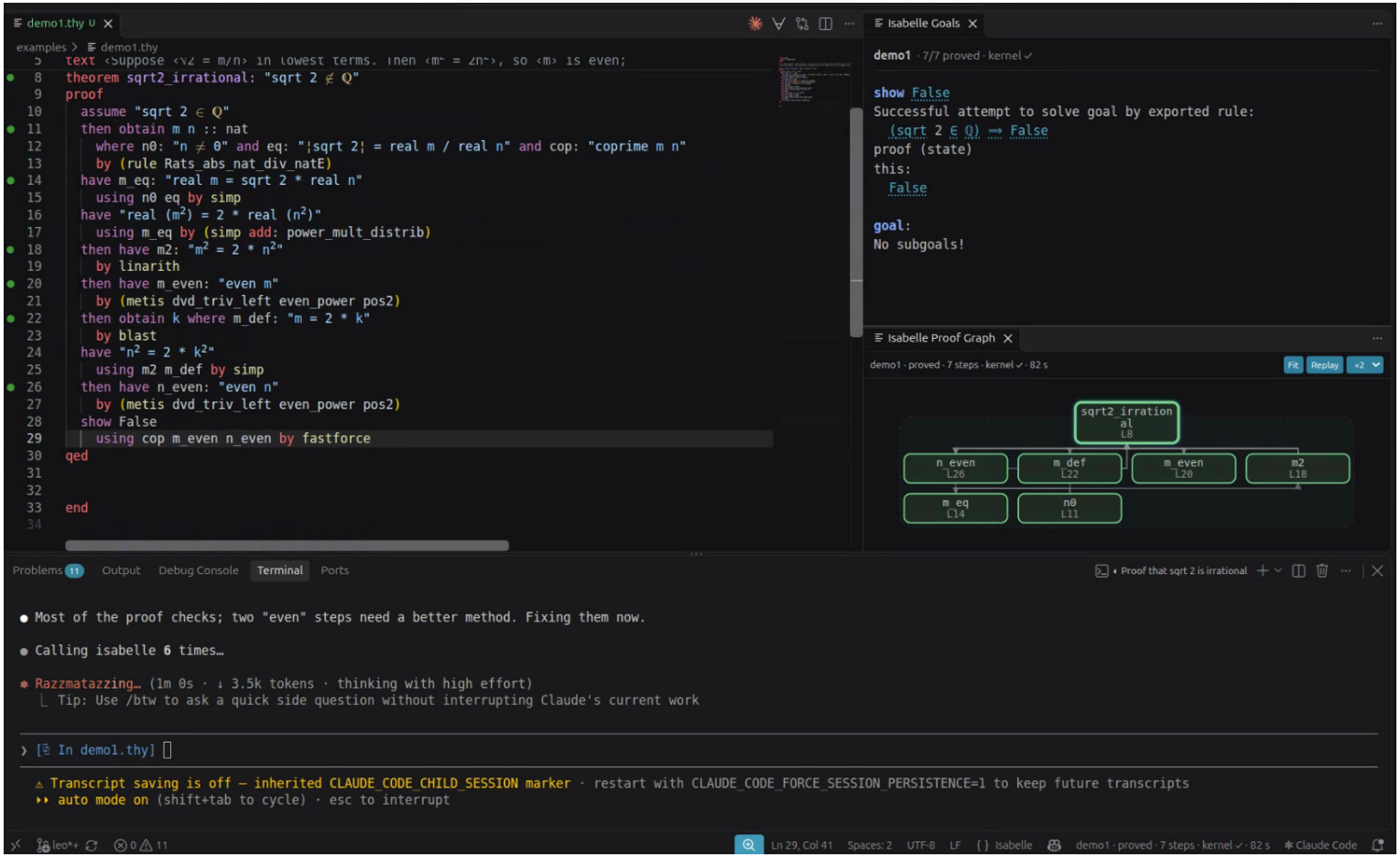


Figure: Jacques Saying this Tool Will Piss Off the Gatekeep-y Isabelle Head Dev

Proving a Biology Fact



https://youtu.be/TgqZSpYr9jc